~J Chap 1.

1. Prove that the Cantor set C is totally disconnected and perfect

proof: (1) $\forall x<y\in C$

$\exists k>0$ s.t. $|x-y|>\frac{1}{3^k}$. then x,y will be seperated by two intervals

(2) Given $x\in C$, $\underset{\forall k>0}{\wedge}$ $\exists$ end-point y_k. s.t. $|x-y_k|\le$ length of $C_k=\frac{1}{3^k}$. $\square$

2. (a) Every number in $[0,1]$ has a ternary expansion:

$x=\sum_{k=1}^{\infty}a_k\cdot 3^{-k}$ $a_k=0,1$ or 2. Prove that $x\in C$ iff x has a representation with $a_k=0$ or 2

(b). The Cantor-Lebesgue function $F: C\to[0,1]$ $\sum_{k=1}^{\infty}a_k\cdot3^{-k}\mapsto\sum_{k=1}^{\infty}b_k\cdot2^{-k}$. $b_k=\frac{a_k}{2}$.

where $a_k\in\{0,2\}$. Show that F is well-defined, continuous and $F(0)=0$, $F(1)=1$

(c). Prove that F is surjective.

(d). Extend F to be a function continuously on $[0,1]$.

Proof: (a). Consider $F_0: C\to\{0,2\}^{\mathbb{N}}$ If x belongs to $[0,\frac{1}{3}]$ then $\pi_1\circ F_0(x)=0$

If x belongs to $[\frac{2}{9},\frac{1}{3}]$ then $\pi_2\circ F_0(x)=2$.

...

On every sheaf, we consider whether x belongs to the left hand or the right hand

$\Rightarrow$ the following number will be 0 or 2 $\Rightarrow a_k\in\{0,2\}$.

(b). 1°. Well-defined: Suppose that $x=\sum_{k=1}^{\infty}a_k\cdot3^{-k}=\sum_{k=1}^{\infty}a_k'\cdot3^{-k}$. $a_k,a_k'\in\{0,2\}$.

The minimal number k s.t $a_k\neq a_k'$ let $a_k=0$. $a_k'=2$.

Then $\sum_{m\geq k}a_m\cdot3^{-m}=\sum_{m\geq k+1}a_m\cdot3^{-m}\le\frac{1}{3^k}<\frac{2}{3^k}<\sum_{m\geq k}a_m'\cdot3^{-k}$

contradiction!

2°. Continuouty: $\forall x_0\in C$: $=\sum_{k=1}^{\infty}a_{k_0}\cdot3^{-k}$. $\forall\varepsilon>0$ $\exists N>0$ s.t. $\sum_{k\geq N}^{\infty}\frac{1}{2^{N+1}}<\varepsilon$

$\forall$ $|x-x_0|<\frac{1}{3^{N+1}}$ we have $a_1=a_{1_0}$, ..., $a_N=a_{N_0}$

$\Longrightarrow b_1=b_{1_0},\dots,b_N=b_{N_0}\Rightarrow|F(x)-F(x_0)|<\sum_{k\geq N+1}^{\infty}\frac{1}{2^{N+1}}<\varepsilon$

Thus F is continuous

3°. $F(0)=0$, $F(1)=1$ is trivial.

(c). $\forall y\in[0,1]$: $=\sum_{k=1}^{\infty}b_k\cdot2^{-k}$. let $x=\sum_{k=1}^{\infty}2b_k\cdot3^{-k}$. we have $\begin{cases}F(x)=y\\ x\in C.\end{cases}$ $\square$

$b_k\in\{0,1\}$.

(d). We can make induction to show that $\forall k>0$, the new end-points of C_k. share the same value of $\frac{1}{2^k}$.

$k=1$, $F(\frac{1}{3})=F(\frac{2}{3})=\frac{1}{2}$

Assume k. consider $k+1$, the new end-point of C_{k+1}: $x_{k+1}=x_k+\frac{1}{3^{k+1}}\Rightarrow F(x_{k+1})=\frac{1}{2^{k+1}}$

7. Cantor-like sets.

Construct a closed set $\hat{C}$ so that at the k^{th} stage of the construction, one removes 2^{k-1} centrally situated open intervals each of length, with $l_1 + 2l_2 + \dots + 2^{k-1}l_k < 1$

(a). If l_j are chosen small enough, s.t. $\sum_{k=1}^{\infty} 2^{k-1} l_k < 1$. Then show that $m(\hat{C}) > 0$

(b) Show that if $x \in \hat{C}$, then $\exists \{x_n\}$ s.t. $x_n \notin \hat{C}$, yet $x_n \to x$, $x_n \in I_n$, where I_n is a sub-interval in the complement of $\hat{C}$ with $|I_n| \to 0$

(c) Prove as a consequence: $\hat{C}$ is perfect, and contains no open interval

(d) $\hat{C}$ is uncountable.

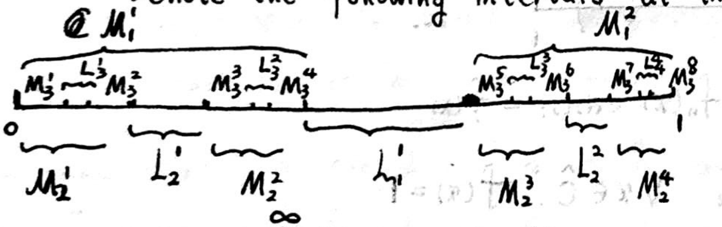
Proof: (a). Denote the ~~denot~~ the delected intervals at the k^{th} stage: $L_k^1, \dots, L_k^{2^{k-1}}$.

Then $\hat{C} = \bigcap_{k=1}^{\infty} [0,1] \setminus (L_k^1 \cup \dots \cup L_k^{2^{k-1}})$. is a ~~Bad~~ closed set $\Rightarrow$ $\hat{C}$ is measurable.

$\Rightarrow m(\hat{C}) = 1 - m\left(\bigcup_{k=1}^{\infty} (L_k^1 \cup \dots \cup L_k^{2^{k-1}})\right) = 1 - \sum_{k=1}^{\infty} 2^{k-1} l_k$.

(b). Obviously, $l_k \to 0$ as k tends to infinity.

Denote the following intervals at the k^{th} stage; $M_k^1, \dots, M_k^{2^k}$.



Given $x \in \hat{C}$ we construct the desired sequence.

Then $\hat{C} = \bigcap_{k=1}^{\infty} (M_k^1 \cup \dots \cup M_k^{2^k})$ $\forall k \geq 1$. $\exists\ 1 \leq j \leq 2^k$. s.t. $x \in M_k^j$.

~~Choose~~ $\exists L_{k+1}^i \subseteq M_k^j$. select $x_k \in L_{k+1}^i$. then $x_k \notin \hat{C}$. $|x_k - x| < |M_k^j|$

$$= \left(1 - \sum_{r=1}^{k-1} 2^{r-1} l_r\right) \frac{1}{2^k}$$

Then we have a sequence $\{x_n\}_{n\geq 2}$. $\begin{cases} |x_n - x| \to 0 & (n \to \infty) \\ x_n \notin \hat{C}. \end{cases}$

$x_n \in I_n$. with $|I_n| \to 0$

(c) (i) Perfectness we have proved in (b).

(ii) distonnectedness is obvious

(d) There is a homeomorphism between $\hat{C}$ and C. $\square$.

9. Give an example of an open set O, such that $\partial(\bar{O})$ has tho positive measure.

Proof: Choose $O = \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{2^{k-1}} L_k^j$ then $\bar{O} = [0,1]$.

$$\Rightarrow m(\partial \bar{O}) = 1 - \sum_{k=1}^{\infty} 2^{k-1} l_k > 0 \ \square$$

10. Construct a decreasing sequence of positive continuous functions on $[0,1]$, whose limit is

not Riemann integrable.

Proof: Let $\hat{C}$ be the Cantor-like set. $F_1(x) = \begin{cases} 1 & x \in [0,1] \setminus L_1^1 \\ \text{a broken line that equals 0 at the cen} & x \in L_1^1 \end{cases}$

$$F_2(x) = \begin{cases} 1 & x \in [0,1] \setminus (L_2^1 \cup L_2^2) \\ \text{a broken lines that equals 0 at the center of the intervals} & x \in L_2^1 \\ & x \in L_2^2 \end{cases}$$

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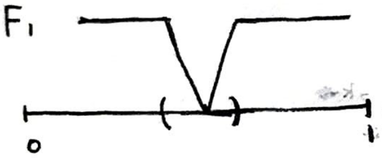
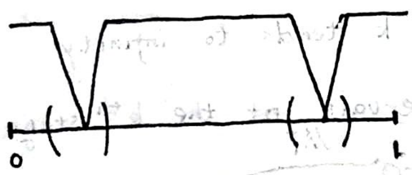
let $f_n = F_1 \cdot F_2 \cdot \ldots \cdot F_n$

F_1



F_2



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Then $0 \le f_n(x) \le 1$, $f_n(x) \ge f_{n+1}(x) \Rightarrow \lim_{n\to\infty} f_n(x)$ exists. $= f(x)$

However: f is discontinuous on $\hat{C}$. Since $\forall x \in \hat{C}$. $f(x) = 1$

By the prop in 4, we can construct a sequence $\{x_n\}_{n\ge1}$. $x_n \notin \hat{C}$ $x_n \to x$

W.L.O.G. x_n is selected at the center of the inter~~g~~val I_n $\Rightarrow f(x_n) = 0$

Contradiction!

Thus f has uncountable discontinuouties $\Rightarrow$ f isn't Riemann integrable $\square$.